Foundation of Mathematics

2- Summary for Limits and Continuity

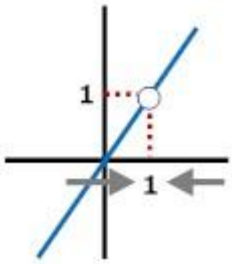
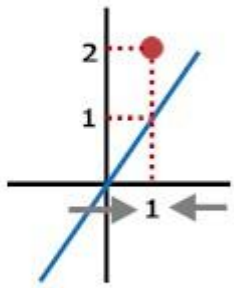
Limits $\lim_{x\to a} f(x) = L$

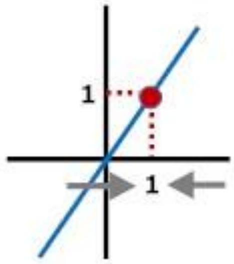
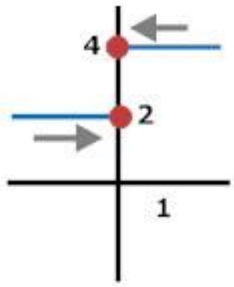
The limit **f(x)** equals **L** if:

- The values of **f(x)** are close to a unique number **L**.
- When the values of **x** are close to a on either side, but not necessarily equal to **a**.

$$\boxed{\lim_{x\to a} f(x) = L \ \text{ OR } \ f(x) \to L \text{ as } x \to a}$$

Examples:

$f(x) = \frac{x^2 - x}{x - 1}$ $f(x) = \frac{x(x-1)}{x-1}$ $f(x) = x; \ x \neq 1$  **f(x) isn't defined at x=1** $\lim_{x\to 1} f(x) = 1$	$f(x) = \begin{cases} \frac{x^2 - x}{x - 1}, & x \neq 1 \\ 2, & x = 1 \end{cases}$ $f(x) = \begin{cases} x; & x \neq 1 \\ 2; & x = 1 \end{cases}$  **f(x) is defined differently at x=1** $\lim_{x\to 1} f(x) = 1$

$f(x) = x$  **f(x) is defined at x=1** $\lim_{x\to 1} f(x) = 1$	$f(x) = \frac{3x + \lvert x\rvert}{x}$ $f(x) = \begin{cases} \frac{3x + x}{x}; & x > 0 \\ \frac{3x - x}{x}; & x < 0 \end{cases}$ $f(x) = \begin{cases} 4 & ; x > 0 \\ 2 & ; x < 0 \end{cases}$  **f(x) isn't defined at x=0** From left: $\lim_{x\to 0^-} f(x) = 2$ From right: $\lim_{x\to 0^+} f(x) = 4$

One sided Limit: $\begin{cases} \text{LEFT}: & \lim_{x\to a^-} f(x) = L \\ \text{RIGHT}: & \lim_{x\to a^+} f(x) = M \end{cases}$

We say that $\lim_{x\to a} f(x)$ exists = **K iff:**

$$\lim_{x\to a^-} f(x) = K = \lim_{x\to a^+} f(x)$$

When to study left and right side limits:

- If f(x) is a **piecewise function** (Defined Differently). $f(x) = \begin{cases} 3x - 2; & x > 2 \\ x; & x < 2 \end{cases}$
- If f(x) contains a modulus. $f(x) = \frac{\lvert x\rvert}{x}$ and the absolute-symmetric point is the limit point.

Techniques of limits:

1. Direct Substitution
2. The limit = $\frac{0}{0}$ = undetermined ,then we must cancel the zero factor "identifying common factors" by:

   i) Factorizing both numerator and denominator.

   ii) Multiplying by the conjugate.
3. If $f(x) = \frac{\sin(x)}{(x)} \Rightarrow \lim_{x\to 0} \frac{\sin(x)}{(x)} = 1$
4. Sandwich Theorem:

   if for $\lim_{x\to c} f(x) = ?$ & $f :: g(x) \le f(x) \le h(x)$

   and $\lim_{x\to c} g(x) = \lim_{x\to c} h(x) = L$

   then $\lim_{x\to c} f(x) = L$

   Use the following:

$$\boxed{\begin{array}{c} -1 \le \sin(?) \le 1 \\ 0 \le \sin^2(?) \le 1 \\ -1 \le \cos(?) \le 1 \\ 0 \le \cos^2(?) \le 1 \\ -1 < \tanh(?) < 1 \\ -\frac{\pi}{2} < \tan^{-1}(?) < \frac{\pi}{2} \end{array}}$$

5. Limits at infinity $\lim_{x\to\infty \text{ or} -\infty} f(x)$

   Divide the numerator and the denominator by the **highest order** term in the **denominator**

- $\frac{\text{Any definite number}\neq 0}{\pm\infty} = \text{zero},$
- **But** $\frac{\infty}{\infty}$= **undetermined** **like** $\frac{0}{0}$ = **Undetermined.**
- **0*(any definite No.)= 0, but**

  $0.(\pm\infty)$=**undetermined**
- $a^{\infty} = \begin{cases} \infty & ; a > 1 \\ 0 & ; a < 1 \\ \text{undetermined} & ; a = 1 \end{cases}$
- $a^{-\infty} = \frac{1}{a^{\infty}}$

6. $\lim \; f(x) = \frac{1}{0}$ **has two meanings:**

   i- **DNE.**

   ii- $+\infty$ **OR** $-\infty$

Remarks:

- ✓ In case we study $\lim_{x\to\pm\infty} \frac{P(x)}{Q(x)}$, but P(x) or Q(x) is not a polynomial, we divide by the **term with highest value** "**producing** $+\infty$" in the denominator.

Ex: $\lim_{x\to-\infty} \frac{e^x - e^{-x}}{e^x + e^{-x}} = \frac{0-\infty}{0+\infty} = -\frac{\infty}{\infty}$, we divide by (e^{-x})

$= \lim_{x\to-\infty} \frac{e^{2x}-1}{e^{2x}+1} = \frac{0-1}{0+1} = -1$

- ✓ We know : $\sqrt{x^2} = |x|$
  1. If we study $\lim\limits_{x\to+\infty}$ and we need to divide by x, here " $\sqrt{x^2} = x$ ".
  2. If we study $\lim\limits_{x\to-\infty}$ and we need to divide by x, here " $\sqrt{x^2} = -x$ ".

Ex: $\lim_{x\to-\infty} \frac{\sqrt{3x^2+x}}{x}$, we divide by the highest power in DENO. : $-x = \sqrt{x^2}$

$= \lim_{x\to-\infty} \frac{\sqrt{3+\frac{1}{x}}}{-1} = -\sqrt{3}$

Continuity at a point

A function f(x) is continuous at x=c if and only if it meets the following three conditions:

1. $f(c)$ **is defined.**
2. $\lim_{x\to c} f(x)$ **exists.**
3. $\lim_{x\to c} f(x) = f(c)$

Note: Any polynomial function is continuous on $\mathbb{R}$.

Types of Discontinuity

if f(x) is not continuous @ *x=a*, we say that "*x=a*" is a discontinuity point of *f*.

Either f(a) is defined or not , We check for:

$$\lim_{x\to a} f(x)$$

If
$\lim_{x\to a} f(x) =$ **exists**
$\Rightarrow @x = a$

f(x) has a
Removable Discontinuity

$$f(x) = \begin{cases} \dfrac{x^2 - x}{x-1} & , x \neq 1 \\ 2 & , x = 1 \end{cases}$$

$$f(x) = \begin{cases} x & ; x \neq 1 \\ 2 & ; x = 1 \end{cases}$$

$$\lim_{x\to 1} f(x) = 1$$

f(1)=2

If $\lim_{x\to a^+} f(x) =$ **L "number"**
&
If $\lim_{x\to a^-} f(x) =$ **M "number"**

and L ≠ M
$\Rightarrow @x = a$

f(x) has a
Jump Discontinuity

$$f(x) = \frac{3x + |x|}{x}$$

$$f(x) = \begin{cases} \dfrac{3x + x}{x} & ; x > 0 \\ \dfrac{3x - x}{x} & ; x < 0 \end{cases}$$

$$f(x) = \begin{cases} 4 & ; x > 0 \\ 2 & ; x < 0 \end{cases}$$

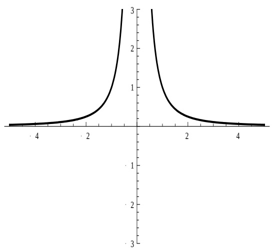
$$\lim_{x\to 0^-} f(x) = 2$$

$$\lim_{x\to 0^+} f(x) = 4$$

If one of the limits
$\lim_{x\to a^+} f(x)$ **OR** $\lim_{x\to a^-} f(x)$

equals ±∞
$\Rightarrow @x = a$

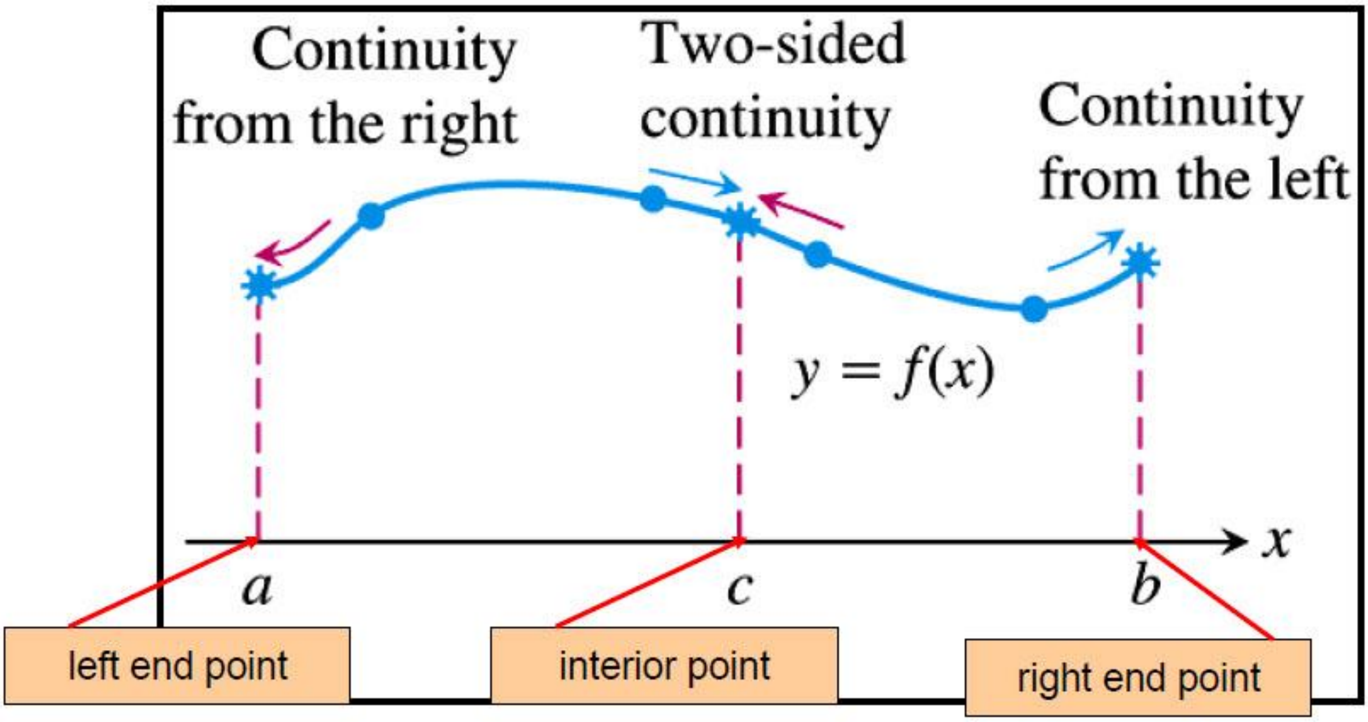
f(x) has an
Infinite Discontinuity



$$\lim_{x\to 0^-} f(x) = \infty$$

$$\lim_{x\to 0^+} f(x) = \infty$$

A function is continuous on a closed interval if is continuous at all interior points as well as continuous at the two end points:



I.V.T "Intermediate Value Theorem"

Given #1: if f is a continuous function on [a, b].
Given #2: if f (a), f (b) with different signs.

$\Longrightarrow$ **Conclusion:** There is a point $c \in (a,b)$ such that

$$f(c) = 0 \;\; \ldots \;\; \text{i.e.,} \begin{cases} \text{c is an x} - \text{intercept for } f(x) \\ \textbf{OR} \\ \text{c is a solution for the equation } f(x) = 0 \end{cases}$$
